

Saachi Agrawal

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Education

The University of Texas at Austin

Expected May 2028

Bachelor of Science in Computer Science, Turing Scholars Honors Program

Bachelor of Science in Mathematics

GPA: 3.80 / 4.0

Relevant Coursework: Honors Data Structures, Honors Discrete Math, Honors Vector Calculus, Honors Computer Architecture, Honors Operating Systems, Honors Algorithms, Linear Algebra, Probability, Differential Equations, Scientific Computation in Numerical Analysis, Artificial Intelligence, Competitive Programming, Number Theory

Technical Skills

Languages: Java, Python, C, C++, JavaScript, HTML, CSS, ARM-64 Assembly, x86-64 Assembly, Verilog, SQL

Tools & Libraries: Git, GitHub, GitHub Copilot, GDB, Bash, Linux, C Standard Library, JUnit, Matplotlib, NumPy

Experience

Applied Research Laboratories, Incoming Intern – Austin, TX

Jun 2026 – Aug 2026

- Accepted offer for the Honors Independent Research and Development Summer Internship Program at ARL:UT
- Anticipated project is Lock-Free Asynchronous Communications Framework under the Space and Geophysics Laboratory

Jane Street INSIGHT, SWE Participant – New York City, NY

Aug 2025

- Built mini projects in OCaml to explore functional programming concepts under mentorship from full-time developers
- Engaged with experts on crucial components of software engineering, including code quality and modularity
- Collaborated and brainstormed solutions in simulations on network protocol transmission and incident response plans

City of Austin, Intern – Austin, TX

Jun 2023 – Jul 2023

- Developed a project that coordinates food donations from restaurants to homeless shelters
- Built registration and login pages using MySQL, Node.js, and Express; rendered pages using JavaScript, HTML, and CSS
- Worked in a group of four, collaborated effectively to create a working prototype from scratch in 7 weeks
- Presented final product to entrepreneurs and developers at the Capital Factory in Austin, Texas

Personal Projects

OS Kernel | C++, x86-64

Aug 2025 – Dec 2025

- Engineered a preemptible, multi-threaded operating system kernel supporting user programs via syscalls, ext2 file system access, and virtual memory
- Enforced safe preemption by disabling interrupts during critical sections and context switches to prevent race conditions
- Built page directories and tracked virtual memory areas (VMAs) to ensure memory isolation between processes
- Designed a Unix-like shell supporting interactive commands & shell scripts, PATH-based executable lookup, and process creation via `fork()` and `exec1()`
 - Validated user input by detecting invalid commands and arguments to prevent erroneous execution
 - Collaborated in a group using Git/GitHub and code review practices to deliver a functional shell within one week

"Fun" Compiler | C, x86-64

Feb 2025 – Apr 2025

- Built a compiler for a toy programming language ("fun") supporting functions, conditionals, and arithmetic operations
- Parsed "fun" syntax into an abstract syntax tree (AST) and used depth-first traversal to generate x86-64 instructions
- Implemented constant folding and tail-call elimination optimizations, reducing compile times to under 0.3 seconds

Web Crawler & Search Engine | Java

Nov 2024 – Dec 2024

- Developed a web crawler using attoparser and Java Standard Library data structures to index ~10,000 pages in 5 minutes, efficiently gathering and storing webpage information in a database
- Implemented an efficient query parsing and search system supporting AND, OR, NOT, and phrases operators to retrieve relevant web pages in seconds
- Tested functionality and scalability using rigorous automated testing with JUnit and randomly generated webs

What's Up With the Weather? | JavaScript, HTML, CSS

Aug 2021 – Nov 2023

- Created a website to educate users on recent climate change research and foster critical thinking on environmental issues
- Designed "playable simulations" using HTML Canvas to engage users with visual learning tools and game-like graphics
- Developed algorithms to model physical processes and mathematical phenomena in nature such as the greenhouse effect
- Compiled academic articles on climate change, simplifying knowledge into accessible content with colloquial language